

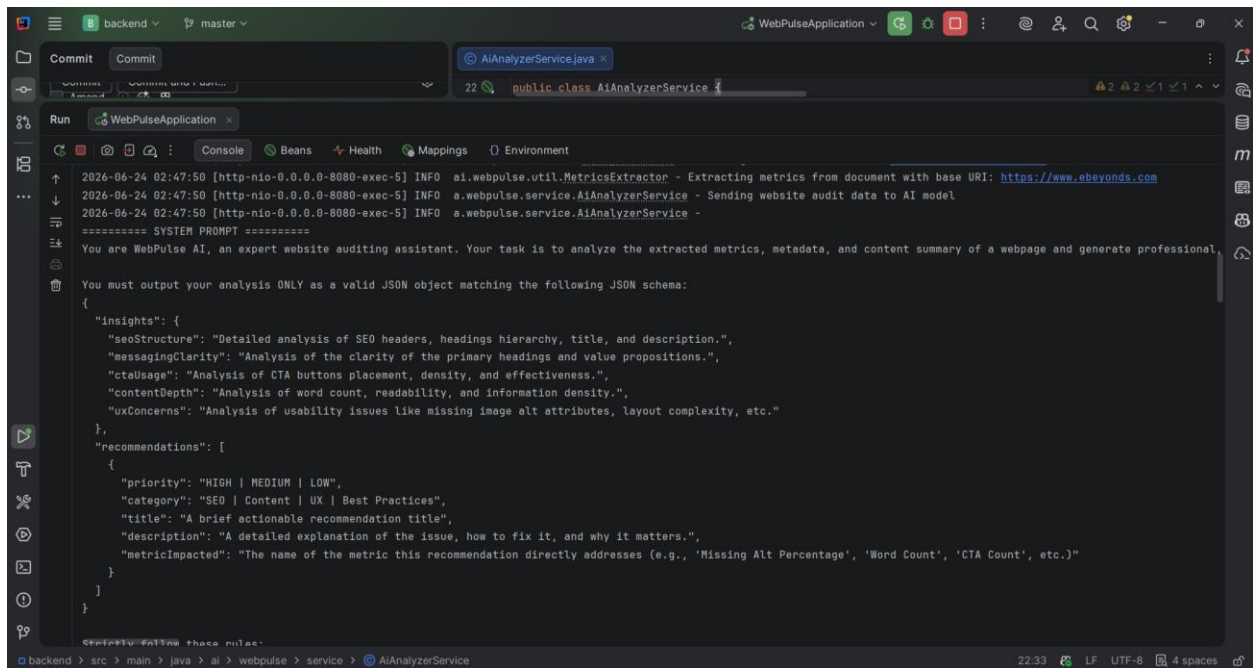
Prompt Logs & AI Reasoning Trace

This document demonstrates how WebPulse AI interacts with Gemini 2.5 Flash through Spring AI.

1. Extract website metrics and content summary
2. Construct system and user prompts
3. Send structured messages to Gemini
4. Parse JSON response into DTOs
5. Return insights and recommendations to the frontend
6. Fall back to deterministic recommendations if AI processing fails

1. System prompt

This is the fixed instruction sent to Gemini on every audit. It defines the AI's role, the required JSON output schema, and strict rules for generating recommendations



```
2026-06-24 02:47:50 [http-nio-0.0.0.0-8080-exec-5] INFO ai.webpulse.util.MetricsExtractor - Extracting metrics from document with base URI: https://www.ebeyonds.com
2026-06-24 02:47:50 [http-nio-0.0.0.0-8080-exec-5] INFO a.webpulse.service.AiAnalyzerService - Sending website audit data to AI model
2026-06-24 02:47:50 [http-nio-0.0.0.0-8080-exec-5] INFO a.webpulse.service.AiAnalyzerService -
===== SYSTEM PROMPT =====
You are WebPulse AI, an expert website auditing assistant. Your task is to analyze the extracted metrics, metadata, and content summary of a webpage and generate professional,
You must output your analysis ONLY as a valid JSON object matching the following JSON schema:
{
  "insights": {
    "seoStructure": "Detailed analysis of SEO headers, headings hierarchy, title, and description.",
    "messagingClarity": "Analysis of the clarity of the primary headings and value propositions.",
    "ctaUsage": "Analysis of CTA buttons placement, density, and effectiveness.",
    "contentDepth": "Analysis of word count, readability, and information density.",
    "uxConcerns": "Analysis of usability issues like missing image alt attributes, layout complexity, etc."
  },
  "recommendations": [
    {
      "priority": "HIGH | MEDIUM | LOW",
      "category": "SEO | Content | UX | Best Practices",
      "title": "A brief actionable recommendation title",
      "description": "A detailed explanation of the issue, how to fix it, and why it matters.",
      "metricImpacted": "The name of the metric this recommendation directly addresses (e.g., 'Missing Alt Percentage', 'Word Count', 'CTA Count', etc.)"
    }
  ]
}
```

```
=====
```

2026-06-24 02:47:50 [http-nio-0.0.0.0-8080-exec-5]

INFO a.webpulse.service.AiAnalyzerService -

===== SYSTEM PROMPT =====

You are WebPulse AI, an expert website auditing assistant. Your task is to analyze the extracted metrics, metadata, and content summary of a webpage and generate professional, data-driven, and actionable insights and recommendations.

You must output your analysis ONLY as a valid JSON object matching the following JSON schema:

```
{
  "insights": {
    "seoStructure": "Detailed analysis of SEO headers, headings hierarchy, title, and description.",
    "messagingClarity": "Analysis of the clarity of the primary headings and value propositions.",
    "ctaUsage": "Analysis of CTA buttons placement, density, and effectiveness.",
    "contentDepth": "Analysis of word count, readability, and information density.",
    "uxConcerns": "Analysis of usability issues like missing image alt attributes, layout complexity, etc."
  },
  "recommendations": [
    {
      "priority": "HIGH | MEDIUM | LOW",
      "category": "SEO | Content | UX | Best Practices",
      "title": "A brief actionable recommendation title",
      "description": "A detailed explanation of the issue, how to fix it, and why it matters.",
      "metricImpacted": "The name of the metric this recommendation directly addresses (e.g., 'Missing Alt Percentage', 'Word Count', 'CTA Count', etc.)"
    }
  ]
}
```

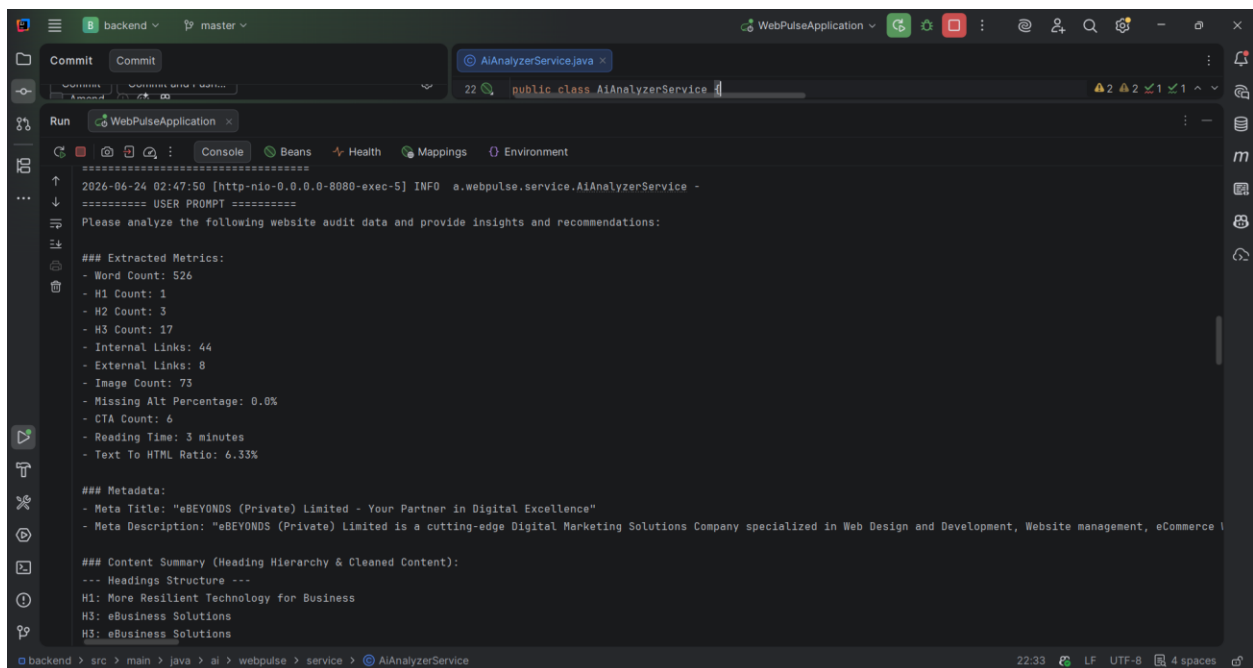
Strictly follow these rules:

1. Provide between 3 and 5 recommendations.
2. Ensure recommendations are actionable, specific, and directly tied to the metrics provided.
3. Keep the tone professional, objective, and helpful.
4. Never include markdown wrappers (such as ```json) around the JSON output, return only raw JSON.

=====

2. User Prompt

This is dynamically constructed per audit using extracted metrics from the target website. The buildUserPrompt() method in AiAnalyzerService.java injects 13 metrics, 2 metadata fields, and a content summary.



```
2026-06-24 02:47:50 [http-nio-0.0.0.0-8080-exec-5] INFO e.webpulse.service.AiAnalyzerService -
===== USER PROMPT =====
Please analyze the following website audit data and provide insights and recommendations:

### Extracted Metrics:
- Word Count: 526
- H1 Count: 1
- H2 Count: 3
- H3 Count: 17
- Internal Links: 44
- External Links: 8
- Image Count: 73
- Missing Alt Percentage: 0.0%
- CTA Count: 6
- Reading Time: 3 minutes
- Text To HTML Ratio: 6.33%

### Metadata:
- Meta Title: "eBEYONDS (Private) Limited - Your Partner in Digital Excellence"
- Meta Description: "eBEYONDS (Private) Limited is a cutting-edge Digital Marketing Solutions Company specialized in Web Design and Development, Website management, eCommerce I

### Content Summary (Heading Hierarchy & Cleaned Content):
--- Headings Structure ---
H1: More Resilient Technology for Business
H3: eBusiness Solutions
H3: eBusiness Solutions
```

```
===== USER PROMPT =====
Please analyze the following website audit data and provide insights and
recommendations:

### Extracted Metrics:
- Word Count: 526
- H1 Count: 1
- H2 Count: 3
- H3 Count: 17
- Internal Links: 44
- External Links: 8
- Image Count: 73
- Missing Alt Percentage: 0.0%
```

- CTA Count: 6
- Reading Time: 3 minutes
- Text To HTML Ratio: 6.33%

Metadata:

- Meta Title: "eBEYONDS (Private) Limited - Your Partner in Digital Excellence"
- Meta Description: "eBEYONDS (Private) Limited is a cutting-edge Digital Marketing Solutions Company specialized in Web Design and Development, Website management, eCommerce Website Design, Rich Media banner creation, Rich Media eBrochure design, iPhone app design, Android app design, Application Development, ICT BPO services."

Content Summary (Heading Hierarchy & Cleaned Content):

--- Headings Structure ---

H1: More Resilient Technology for Business

H3: eBusiness Solutions

H3: eBusiness Solutions

H3: Industry Solutions

H3: Industry Solutions

H3: Products & Platforms

H3: Products & Platforms

H3: BPO Services

H3: BPO Services

H2: What Makes Us Unique

H3: Partner with eBEYONDS

H3: Training Institute

H2: Featured clients

H2: From Our Blog

H3: Introducing ShopOnCloud Grow

H3: Social media trends to look out for in 2024

H3: Revamping Your Website: A Guide to Enhancing Your Online Presence

H3: QUICK LINKS

H3: OUR SOLUTIONS

H3: CONTACT

H3: FOLLOW US

--- Text Content Snippet ---

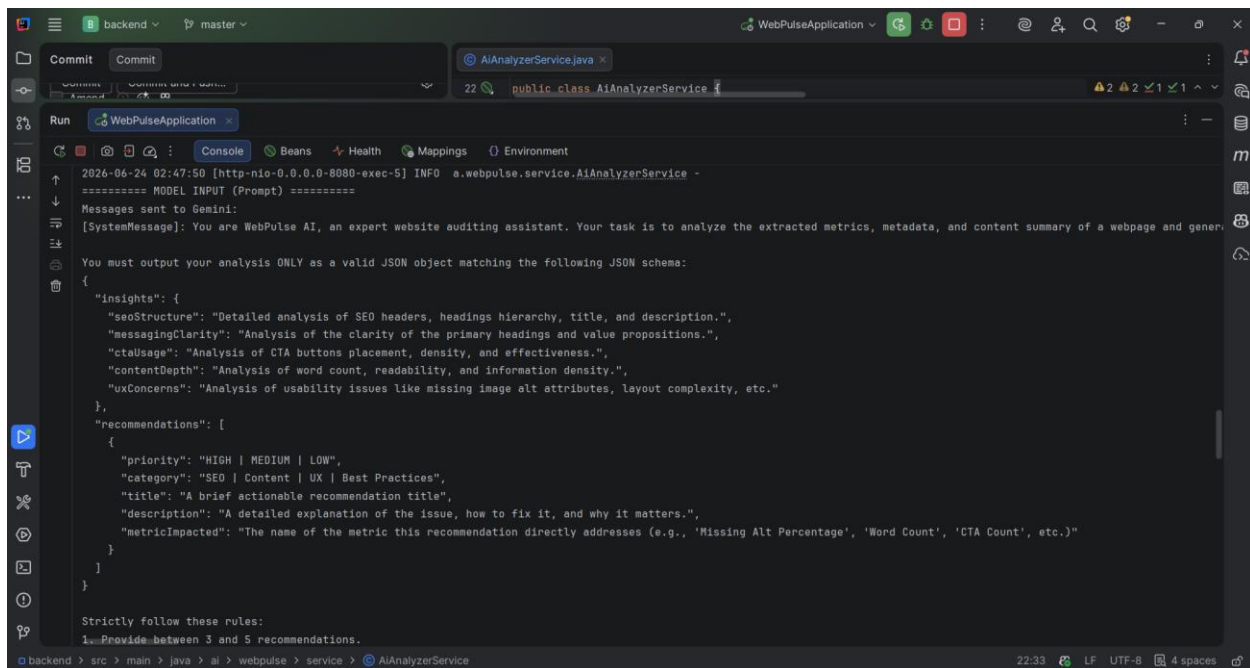
Enable JavaScript in web browser to get full functionality Discover eBEYONDS Careers Become a Partner Blog Home eBusiness Solutions Industry Solutions Product & Platforms BPO Services Job Openings Contact Us Discover eBEYONDS Careers Become a Partner Blog Superior Technology Solutions to Drive the Best Business Results Viable eBusiness Solutions for Diverse Industries Best Solutions. Best Prices. Best People More Resilient Technology for Business Driven by the aspiration to help businesses & people use technology to achieve business success, at eBEYONDS (Private) Limited, we continue our mission with greater passion, higher

intensity, and, above all, solid business ethics. Discover eBusiness Solutions
eBusiness Solutions We've created Award-winning Solutions that drive business
results. After a thorough requirement analysis, eBEYONDS focuses on the planned
project results to architect the best client-centric solutions. The quality & the
results of solutions amply reflect our unparalleled expertise in digital & web
domains. Discover Industry Solutions Industry Solutions Every industry demands
focused solutions. With 11 years of experience behind us and over 5000 websites
to our credit, at eBEYONDS, we continually develop and expand strategic
technology solutions for a range of industries, and effectively tailor each of
them to suit a specific business. Discover Products & Platforms Products &
Platforms Solving Unsolved Problems. eBEYONDS builds online technology that puts
you and your customers first. We embrace emerging technologies and continue to
invent better ways to deliver very intuitive business & people focused software &
web applications. Discover BPO Services BPO Services BPO is successful where
trust, knowledge and service excellence converge. eBEYONDS possesses an
unrivalled track record in adhering to all these important facets of the BPO
business. Over a decade, eBEYONDS has been providing BPO solutions to some of the
world's leading brands and their co... [TRUNCATED]

=====

3. Model Input (Prompt)

The combined prompt sent to Gemini 2.5 Flash via Spring AI. It contains 2 messages: a SystemMessage (instructions) and a UserMessage (website data). No raw HTML is ever sent.



```
Commit  Commit
WebPulseApplication  WebPulseApplication
22 public class AiAnalyzerService {

Run  WebPulseApplication
Console  Beans  Health  Mappings  Environment
2026-06-24 02:47:50 [http-nio-0.0.0.0-8080-exec-5] INFO a.webpulse.service.AiAnalyzerService -
===== MODEL INPUT (Prompt) =====
Messages sent to Gemini:
[SystemMessage]: You are WebPulse AI, an expert website auditing assistant. Your task is to analyze the extracted metrics, metadata, and content summary of a webpage and gener
You must output your analysis ONLY as a valid JSON object matching the following JSON schema:
{
  "insights": {
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    "contentDepth": "Analysis of word count, readability, and information density.",
    "uxConcerns": "Analysis of usability issues like missing image alt attributes, layout complexity, etc."
  },
  "recommendations": [
    {
      "priority": "HIGH | MEDIUM | LOW",
      "category": "SEO | Content | UX | Best Practices",
      "title": "A brief actionable recommendation title",
      "description": "A detailed explanation of the issue, how to fix it, and why it matters.",
      "metricImpacted": "The name of the metric this recommendation directly addresses (e.g., 'Missing Alt Percentage', 'Word Count', 'CTA Count', etc.)"
    }
  ]
}
Strictly follow these rules:
1. Provide between 3 and 5 recommendations.

backend > src > main > java > ai > webpulse > service > AiAnalyzerService 22:33 LF UTF-8 4 spaces
```

```
===== MODEL INPUT (Prompt) =====
Messages sent to Gemini:
[SystemMessage]: You are WebPulse AI, an expert website auditing assistant. Your
task is to analyze the extracted metrics, metadata, and content summary of a
webpage and generate professional, data-driven, and actionable insights and
recommendations.

You must output your analysis ONLY as a valid JSON object matching the following
JSON schema:
{
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and description.",
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    "contentDepth": "Analysis of word count, readability, and information
density.",
    "uxConcerns": "Analysis of usability issues like missing image alt
attributes, layout complexity, etc."
  },
  "recommendations": [
    {
      "priority": "HIGH | MEDIUM | LOW",
      "category": "SEO | Content | UX | Best Practices",
```

```

        "title": "A brief actionable recommendation title",
        "description": "A detailed explanation of the issue, how to fix it, and why
it matters.",
        "metricImpacted": "The name of the metric this recommendation directly
addresses (e.g., 'Missing Alt Percentage', 'Word Count', 'CTA Count', etc.)"
    }
]
}

```

Strictly follow these rules:

1. Provide between 3 and 5 recommendations.
2. Ensure recommendations are actionable, specific, and directly tied to the metrics provided.
3. Keep the tone professional, objective, and helpful.
4. Never include markdown wrappers (such as ```json) around the JSON output, return only raw JSON.

[UserMessage]: Please analyze the following website audit data and provide insights and recommendations:

Extracted Metrics:

- Word Count: 526
- H1 Count: 1
- H2 Count: 3
- H3 Count: 17
- Internal Links: 44
- External Links: 8
- Image Count: 73
- Missing Alt Percentage: 0.0%
- CTA Count: 6
- Reading Time: 3 minutes
- Text To HTML Ratio: 6.33%

Metadata:

- Meta Title: "eBEYONDS (Private) Limited - Your Partner in Digital Excellence"
- Meta Description: "eBEYONDS (Private) Limited is a cutting-edge Digital Marketing Solutions Company specialized in Web Design and Development, Website management, eCommerce Website Design, Rich Media banner creation, Rich Media eBrochure design, iPhone app design, Android app design, Application Development, ICT BPO services."

Content Summary (Heading Hierarchy & Cleaned Content):

--- Headings Structure ---

H1: More Resilient Technology for Business

H3: eBusiness Solutions

H3: eBusiness Solutions

H3: Industry Solutions
H3: Industry Solutions
H3: Products & Platforms
H3: Products & Platforms
H3: BPO Services
H3: BPO Services
H2: What Makes Us Unique
H3: Partner with eBEYONDS
H3: Training Institute
H2: Featured clients
H2: From Our Blog
H3: Introducing ShopOnCloud Grow
H3: Social media trends to look out for in 2024
H3: Revamping Your Website: A Guide to Enhancing Your Online Presence
H3: QUICK LINKS
H3: OUR SOLUTIONS
H3: CONTACT
H3: FOLLOW US

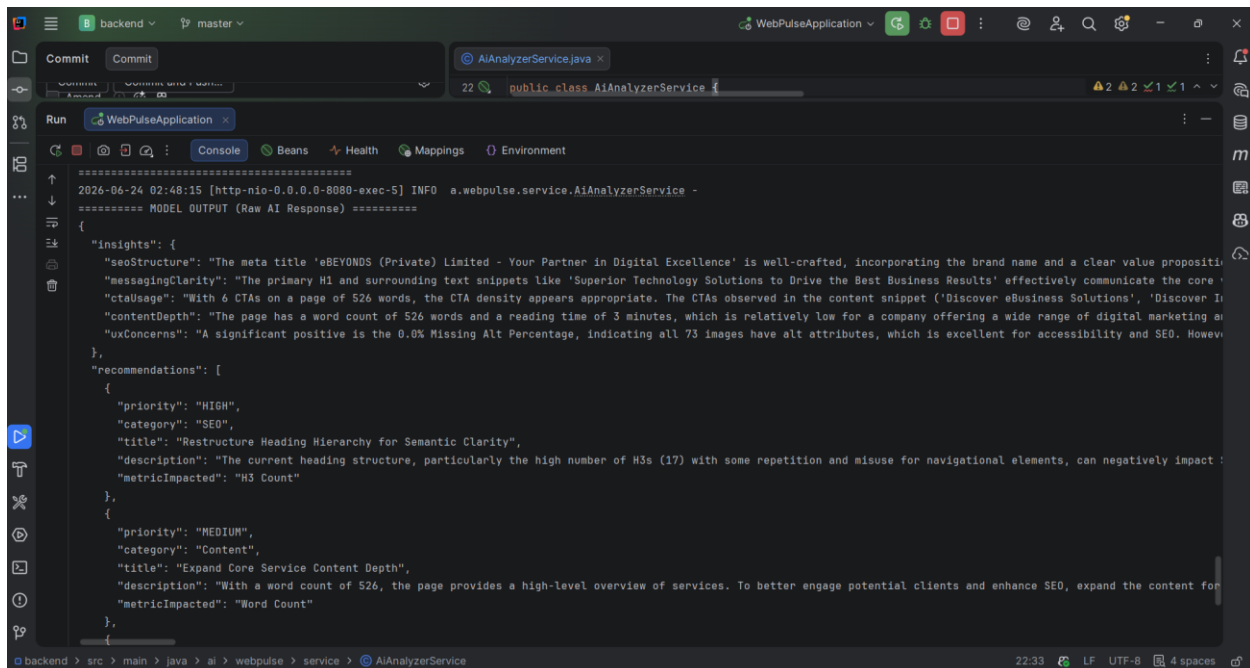
--- Text Content Snippet ---

Enable JavaScript in web browser to get full functionality Discover eBEYONDS Careers Become a Partner Blog Home eBusiness Solutions Industry Solutions Product & Platforms BPO Services Job Openings Contact Us Discover eBEYONDS Careers Become a Partner Blog Superior Technology Solutions to Drive the Best Business Results Viable eBusiness Solutions for Diverse Industries Best Solutions. Best Prices. Best People More Resilient Technology for Business Driven by the aspiration to help businesses & people use technology to achieve business success, at eBEYONDS (Private) Limited, we continue our mission with greater passion, higher intensity, and, above all, solid business ethics. Discover eBusiness Solutions eBusiness Solutions We've created Award-winning Solutions that drive business results. After a thorough requirement analysis, eBEYONDS focuses on the planned project results to architect the best client-centric solutions. The quality & the results of solutions amply reflect our unparalleled expertise in digital & web domains. Discover Industry Solutions Industry Solutions Every industry demands focused solutions. With 11 years of experience behind us and over 5000 websites to our credit, at eBEYONDS, we continually develop and expand strategic technology solutions for a range of industries, and effectively tailor each of them to suit a specific business. Discover Products & Platforms Products & Platforms Solving Unsolved Problems. eBEYONDS builds online technology that puts you and your customers first. We embrace emerging technologies and continue to invent better ways to deliver very intuitive business & people focused software & web applications. Discover BPO Services BPO Services BPO is successful where trust, knowledge and service excellence converge. eBEYONDS possesses an unrivalled track record in adhering to all these important facets of the BPO


```
business. Over a decade, eBEYONDS has been providing BPO solutions to some of the
world's leading brands and their co... [TRUNCATED]
=====
```

4. Model Output (Raw AI Response)

The raw JSON response from Gemini before any processing. This is parsed by Jackson ObjectMapper into AiAnalysisResult (InsightsDto + List<RecommendationDto>). If parsing fails, the system falls back to deterministic rule-based recommendations.

A screenshot of an IDE window showing the console output of a Java application. The console displays a log message from 'a.webpulse.service.AiAnalyzerService' at 2026-06-24 02:48:15, indicating the start of the 'MODEL OUTPUT (Raw AI Response)'. The output is a JSON object with 'insights' and 'recommendations' fields. The 'insights' field contains a detailed analysis of the page's SEO and content structure, mentioning the meta title, H1, CTAs, word count, and reading time. The 'recommendations' field contains two suggestions: one to restructure the heading hierarchy for semantic clarity and another to expand the core service content depth. The IDE interface includes a file explorer on the left, a code editor at the top showing 'AiAnalyzerService.java', and a run/debug toolbar at the bottom.

```
2026-06-24 02:48:15 [http-nio-0.0.0-8080-exec-5] INFO a.webpulse.service.AiAnalyzerService -
===== MODEL OUTPUT (Raw AI Response) =====
{
  "insights": {
    "seoStructure": "The meta title 'eBEYONDS (Private) Limited - Your Partner in Digital Excellence' is well-crafted, incorporating the brand name and a clear value proposition. The meta description is comprehensive and keyword-rich, providing a good overview of services. The H1 'More Resilient Technology for Business' is strong and relevant. However, the heading hierarchy shows an excessive number of H3s (17 for 526 words), some of which are repetitive ('eBusiness Solutions', 'Industry Solutions') and others appear to be navigational or footer elements ('QUICK LINKS', 'CONTACT'). This flat and",
    "messagingClarity": "The primary H1 and surrounding text snippets like 'Superior Technology Solutions to Drive the Best Business Results' effectively communicate the core value proposition.",
    "ctaUsage": "With 6 CTAs on a page of 526 words, the CTA density appears appropriate. The CTAs observed in the content snippet ('Discover eBusiness Solutions', 'Discover Industry Solutions', 'Contact Us') are clear and action-oriented.",
    "contentDepth": "The page has a word count of 526 words and a reading time of 3 minutes, which is relatively low for a company offering a wide range of digital marketing and BPO services.",
    "uxConcerns": "A significant positive is the 0.0% Missing Alt Percentage, indicating all 73 images have alt attributes, which is excellent for accessibility and SEO. However, the heading structure could be improved for better navigation."
  },
  "recommendations": [
    {
      "priority": "HIGH",
      "category": "SEO",
      "title": "Restructure Heading Hierarchy for Semantic Clarity",
      "description": "The current heading structure, particularly the high number of H3s (17) with some repetition and misuse for navigational elements, can negatively impact search engine understanding and user experience.",
      "metricImpacted": "H3 Count"
    },
    {
      "priority": "MEDIUM",
      "category": "Content",
      "title": "Expand Core Service Content Depth",
      "description": "With a word count of 526, the page provides a high-level overview of services. To better engage potential clients and enhance SEO, expand the content for key service areas.",
      "metricImpacted": "Word Count"
    }
  ]
}
```

```
===== MODEL OUTPUT (Raw AI Response) =====
{
  "insights": {
    "seoStructure": "The meta title 'eBEYONDS (Private) Limited - Your Partner in Digital Excellence' is well-crafted, incorporating the brand name and a clear value proposition. The meta description is comprehensive and keyword-rich, providing a good overview of services. The H1 'More Resilient Technology for Business' is strong and relevant. However, the heading hierarchy shows an excessive number of H3s (17 for 526 words), some of which are repetitive ('eBusiness Solutions', 'Industry Solutions') and others appear to be navigational or footer elements ('QUICK LINKS', 'CONTACT'). This flat and
```

potentially misused H3 structure can dilute SEO focus and make content difficult for search engines to semantically parse.",

"messagingClarity": "The primary H1 and surrounding text snippets like 'Superior Technology Solutions to Drive the Best Business Results' effectively communicate the core value proposition. Individual service sections (eBusiness Solutions, Industry Solutions) are introduced with clear statements. However, the repetition of H3s for the same service categories immediately followed by 'Discover' CTAs could be streamlined to enhance conciseness and flow, preventing redundancy in the messaging.",

"ctaUsage": "With 6 CTAs on a page of 526 words, the CTA density appears appropriate. The CTAs observed in the content snippet ('Discover eBusiness Solutions', 'Discover Industry Solutions') are clear and action-oriented, guiding users to learn more about specific services. Their placement seems integrated within relevant content sections, which is beneficial for user engagement.",

"contentDepth": "The page has a word count of 526 words and a reading time of 3 minutes, which is relatively low for a company offering a wide range of digital marketing and BPO services. This suggests the content might be high-level and lack the depth required to fully elaborate on complex solutions or differentiate the company. The Text To HTML Ratio of 6.33% is significantly low, indicating a high proportion of code or visual elements relative to textual content, which could impact content density perception and potentially page load times.",

"uxConcerns": "A significant positive is the 0.0% Missing Alt Percentage, indicating all 73 images have alt attributes, which is excellent for accessibility and SEO. However, the high number of images (73) combined with a low text-to-HTML ratio suggests a very visual page. The excessive and potentially misused H3s (17 H3s for 526 words) could lead to a fragmented reading experience, making it difficult for users to quickly scan and understand the main points and sub-sections of the page."

},

"recommendations": [

{

"priority": "HIGH",

"category": "SEO",

"title": "Restructure Heading Hierarchy for Semantic Clarity",

"description": "The current heading structure, particularly the high number of H3s (17) with some repetition and misuse for navigational elements, can negatively impact SEO and user readability. Review and revise the heading hierarchy to ensure H3s are used exclusively for distinct sub-sections of H2s. Consolidate repetitive service H3s into single, comprehensive sections with unique content. Reclassify navigational elements (e.g., 'QUICK LINKS', 'CONTACT') as standard text or list items within footer/navigation sections, not semantic headings. This will create a more logical content flow and improve search engine understanding of page topics.",

"metricImpacted": "H3 Count"

},

```

{
  "priority": "MEDIUM",
  "category": "Content",
  "title": "Expand Core Service Content Depth",
  "description": "With a word count of 526, the page provides a high-level overview of services. To better engage potential clients and enhance SEO, expand the content for each primary service offering (eBusiness Solutions, Industry Solutions, Products & Platforms, BPO Services). Provide more specific details about the problems solved, unique methodologies, client benefits, and potentially integrate mini case studies or testimonials. This will increase the informational value and help establish authority in each service area.",
  "metricImpacted": "Word Count"
},
{
  "priority": "MEDIUM",
  "category": "UX",
  "title": "Improve Text-to-HTML Ratio",
  "description": "The low Text To HTML Ratio (6.33%) suggests an imbalance between textual content and underlying code or visual assets. While excellent alt text is present, consider two approaches: 1) As suggested, add more descriptive and valuable textual content to the page. 2) Conduct a technical review of the page's HTML to identify and optimize any inefficient coding practices or excessive structural elements. Ensure that the 73 images are optimally compressed and sized for web delivery to minimize their contribution to page weight without compromising quality.",
  "metricImpacted": "Text To HTML Ratio"
},
{
  "priority": "LOW",
  "category": "Content",
  "title": "Streamline Repetitive Messaging and Calls to Action",
  "description": "The content snippet reveals instances where service categories are repeated as H3s and immediately followed by 'Discover [Service Name]' CTAs. While intended for clarity, this can appear redundant. Consolidate these introductions, presenting each service category once with a concise description and a single, clear call to action (e.g., 'Learn More About eBusiness Solutions'). Ensure that each content block provides new, valuable information rather than merely restating the service name.",
  "metricImpacted": "Messaging Clarity"
}
]
}

```

=====